AMADEUS SCHEMA INTEGRATION AT THE STORAGE BOUNDARY

Current release • ordinary scientific calls • post-calculation enforcement

AI: development/review assistance only. Runtime: deterministic code. Human scientific approvals: pending.

01 Build contracts and catalog	Build once per changed release: LinkML modules, Amadeus mapping and frozen code/unit policy/evidence. QA produces pinned JSON/SQL/catalog/registry. dev/check_approval.R; validate_release.R. Scientific approval pending.
	HUMAN: Approve source authority, units, temporal meaning and scope
02 User request	Input: variables, dates, sites and paths. Ordinary Amadeus arguments; no runtime spreadsheet/metadata merge. Integration profile is 16 native daily codes, one-year point extraction.
	HUMAN: Choose scientific question, sites, dates and supported method
03 Amadeus download	download_data(dataset_name="gridmet", ...). Output: source assets or reused cache. Recorder observes source identity; a route call does not prove fresh acquisition.
	HUMAN: Confirm source/version/cache policy and acquisition claims
04 Amadeus process	process_covariates(covariate="gridmet", ...). Output: processed rasters and common TIFF handoff. Caller supplies extent/dates; no new scientific dispatcher.
	HUMAN: Approve scientific processing; review portability changes
05 Amadeus calculate	calculate_covariates(covariate="gridmet", ...). Output: ordinary R result frames. Native daily, radius 0 / mean, no temporal aggregation in this tested profile.
	HUMAN: Choose/approve extraction method and tolerances
06 Verify recorded context	export_gridmet() after calculation verifies request, function/output and recording consistency. Input: calculated/processed outputs and explicit execution context. Local unsigned evidence.
	HUMAN: Review provenance truth, evidence limits and capture completeness
07 Enforce actual-data contract	schema_export()/storage_validate(): source units, expected membership/time labels, finite values, JSON schema, geometry/references/lineage rules. Output: accepted instance or rejection.
	HUMAN: Approve unit authority, time semantics, binding and missingness
08 Typed Parquet and readback	Storage boundary writes/reads typed staging records; shared semantic checks remain. Adapter also materializes Parquet. No footer-only enforcement or zero-copy claim.
	HUMAN: Approve logical types, null semantics and required readback fidelity
09 Load and verify DuckDB	database.R/parquet_experiment.R map/load accepted rows, run 102 SQL queries, check tolerances and reopen. Empty-table checks do not establish full-model coverage.
	HUMAN: Approve key/reference rules, tolerance and coverage interpretation
10 Bind the published snapshot	storage_publish_manifest(): hashes schema/acceptance/results/source metadata, recording, DB/receipt and accepted Parquet. Relative whitelisted paths; release and instance bindings.
	HUMAN: Approve trust model and required protected artifacts
11 Complete and publish pointer	Critical files marked readonly; workflow receipt binds manifest SHA; final LATEST replacement. Failed load retains diagnostics but does not complete/advance pointer. Not multi-file crash durability.
	HUMAN: Approve filesystem/recovery and retention requirements
12 Verify open and query	database.R RUN duckdb SQL invokes storage_verify_publication() once per open, then read-only query. Checks completion/release/file bytes; no source NetCDF reread. Direct IDE/DBI bypasses guards.
	HUMAN: Choose supported access path and deployment controls
13 Separate scientific/replay review	QA direct NetCDF comparison passed 336 values locally. Human unit/physical-time/scope sign-off and independent colleague/environment replay remain pending. Not production approval.
	HUMAN: Approve units, physical time, binding, provenance, acceptance; independently replay; decide release

During stages 3–5, the recorder observes calls, source hashes and outputs; it does not run science or enforce the schema.

Trace authority → implementation → tests → human decision in the supplied registers. No new runtime AI or approval service.